

Pentium 4 ITLB, Branch Prediction & Hyper-Threading -- MPCA Module 6

Definition: Pentium 4 integrates an Instruction Translation Lookaside Buffer (ITLB), advanced Dynamic Branch Prediction, and Hyper-Threading (HT) Technology (Simultaneous Multithreading) to maximize execution unit utilization.

1. Detailed Technical Explanation

1. Instruction Translation Lookaside Buffer (ITLB)

- * A high-speed associative hardware cache that caches virtual-to-physical page address translations for instruction fetch.

2. Advanced Dynamic Branch Prediction

- * Because the NetBurst pipeline is 20 stages deep, a branch misprediction penalty results in flushing 20 stages of instructions (a massive performance penalty).

3. Hyper-Threading Technology (HT Technology)

Hyper-Threading is Intel's hardware implementation of Simultaneous Multithreading (SMT) that allows a single physical processor core to appear as two logical processors to the operating system.

WITHOUT HYPER-THREADING:

Physical Core: [Thread 1] ==> Idle Execution Units (Wasted Silicon)

WITH HYPER-THREADING (Simultaneous Multithreading):

Logical Core 1: [Thread 1] ==> Shared Physical Execution Units (ALU, FPU, Caches)

Logical Core 2: [Thread 2] / (Zero Idle Slots, 15-30% Throughput Boost!)

Architecture of Hyper-Threading:

1. Duplicated Architectural State:

4. Must-Write Points for Exams

- * Hyper-Threading presents one physical core as two logical cores to the OS scheduler.

5. Quick Recall Flow

ITLB Page Caching -> Advanced BTB (>94% Branch Accuracy) -> Hyper-Threading: 1 Physical Core = 2 Logical Processors (Duplicated State, Shared ALUs)